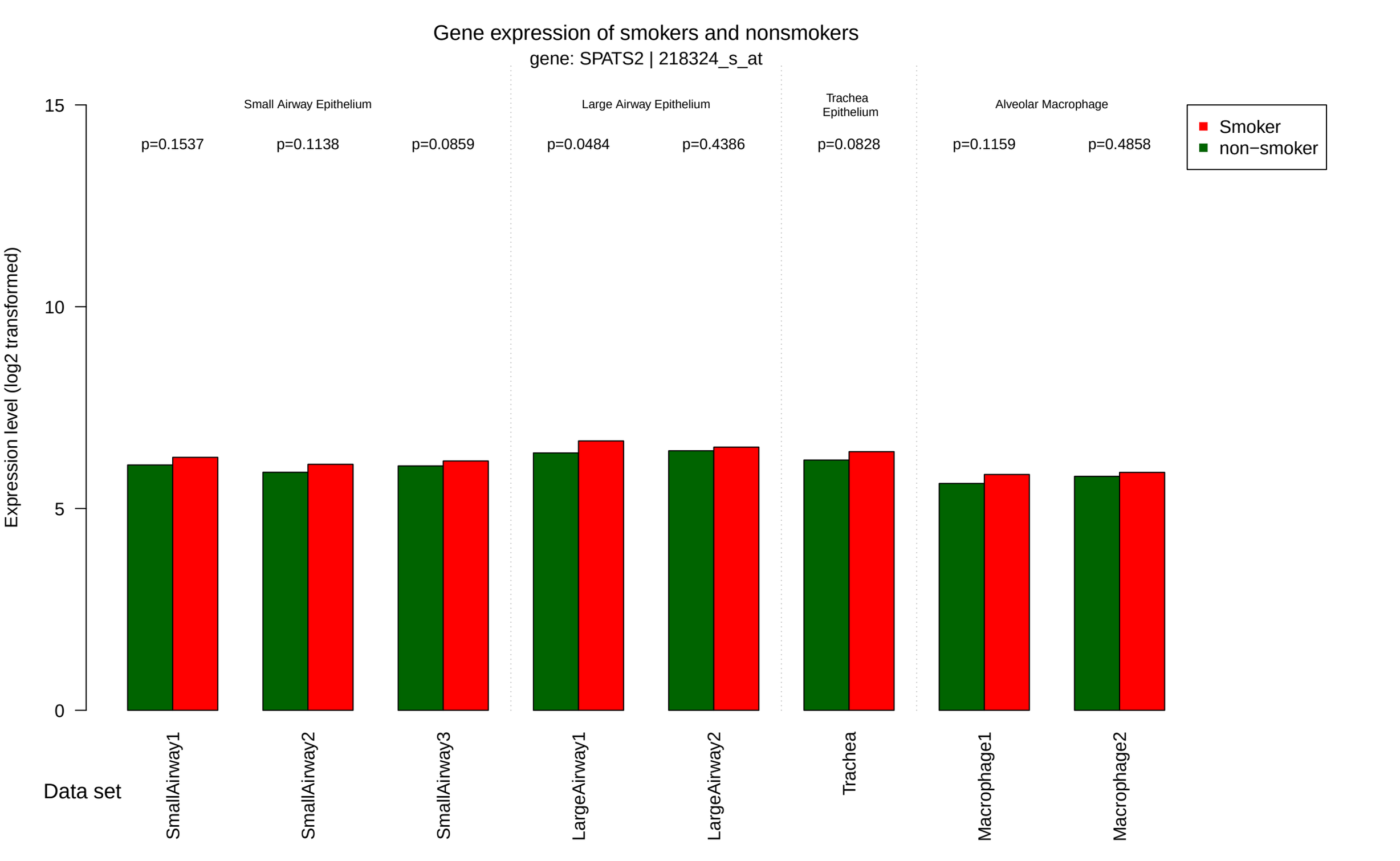
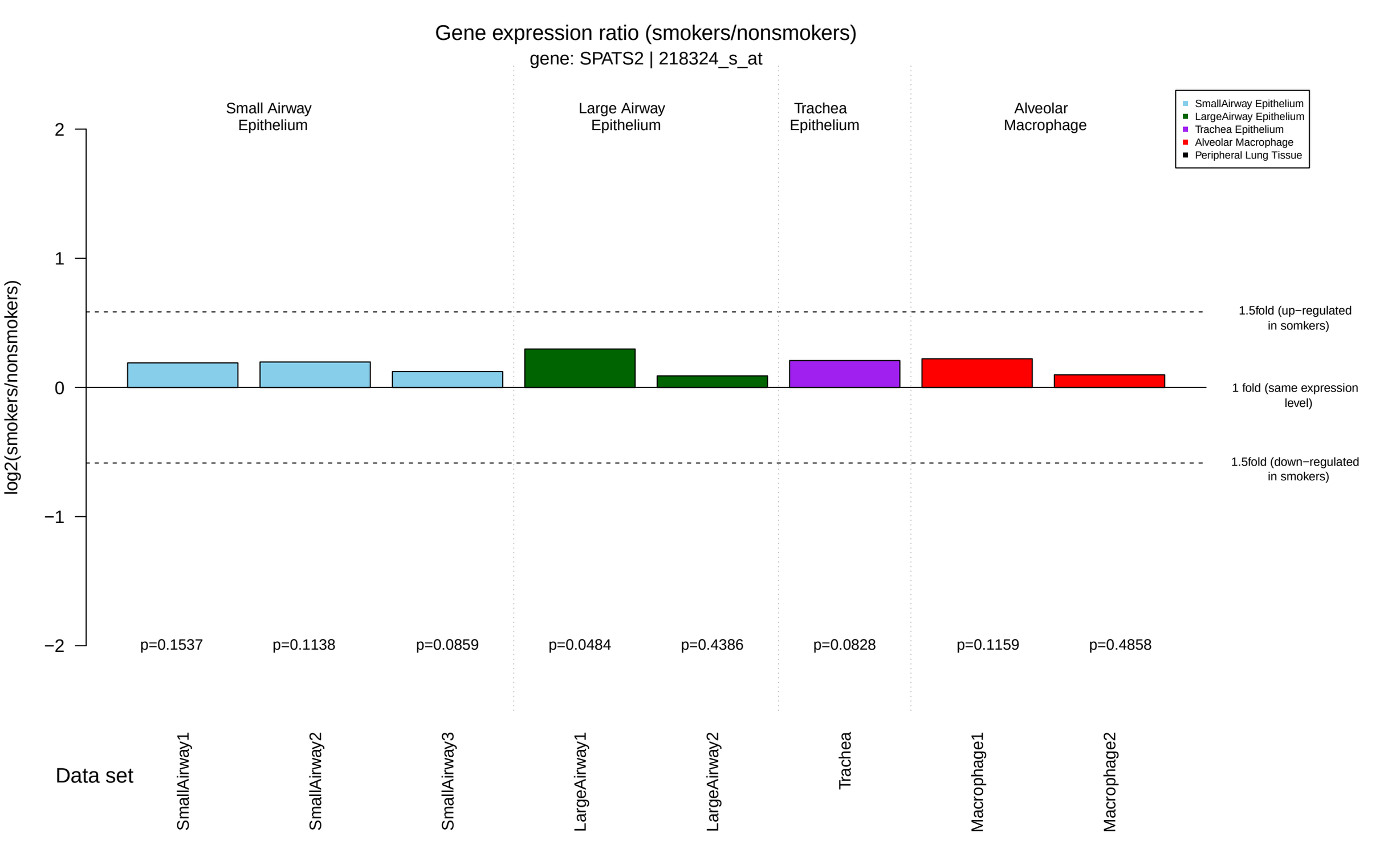
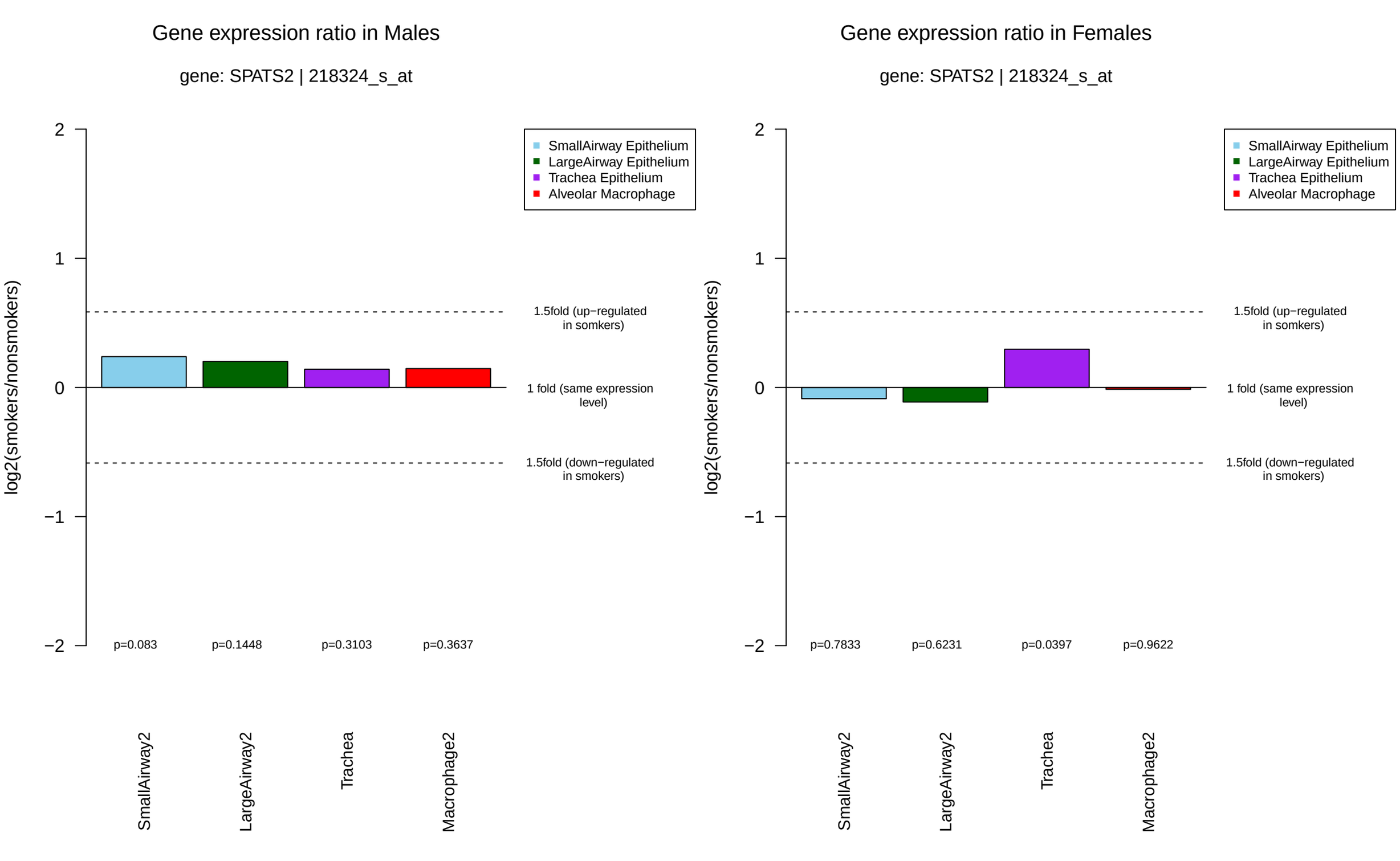
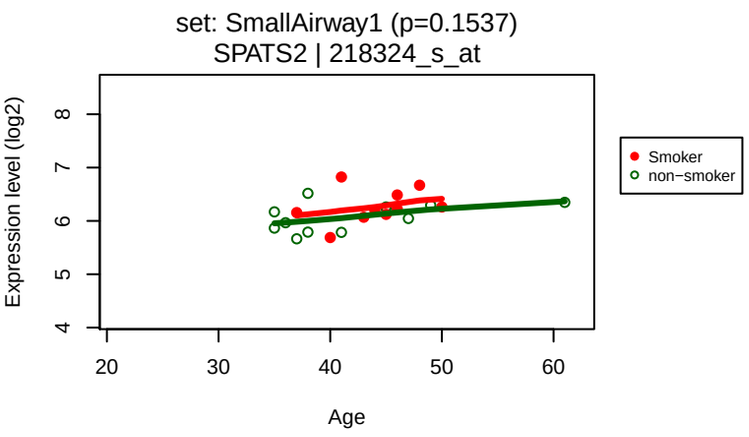
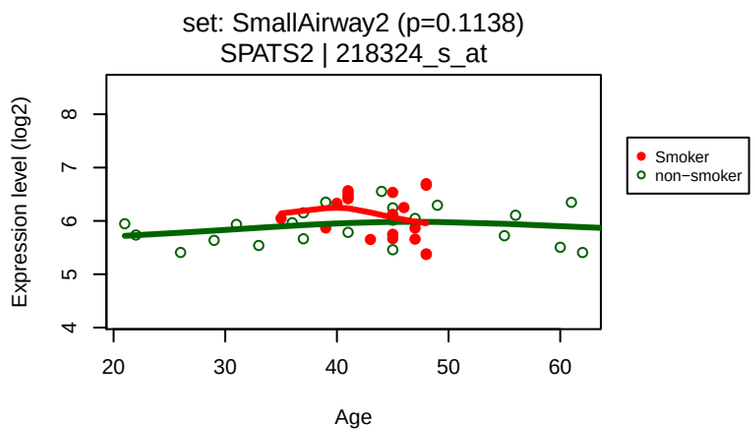
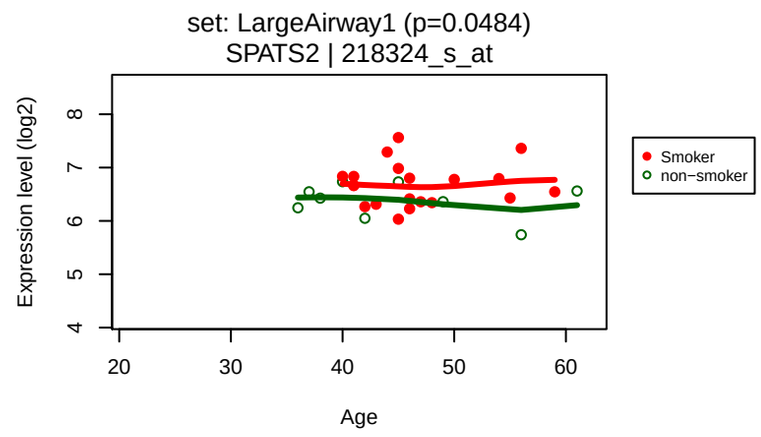

Gene expression of smokers and nonsmokers
gene: SPATS2 | 218324_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1537
p=0.1138
p=0.0859
p=0.0484
p=0.4386
p=0.0828
p=0.1159
p=0.4858
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

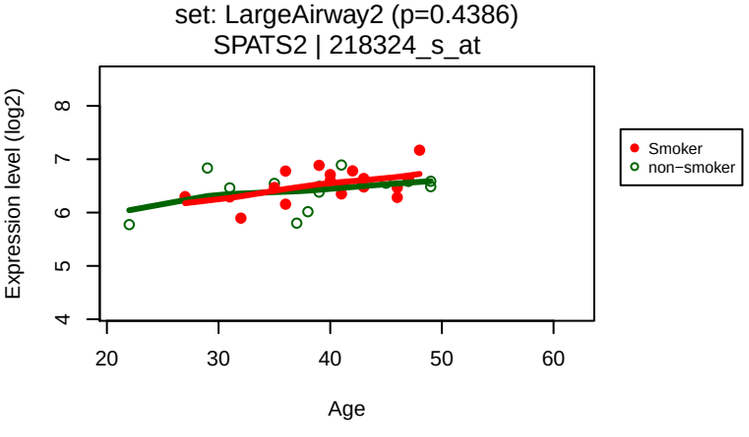

Gene expression ratio (smokers/nonsmokers)
gene: SPATS2 | 218324_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1537
p=0.1138
p=0.0859
p=0.0484
p=0.4386
p=0.0828
p=0.1159
p=0.4858
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

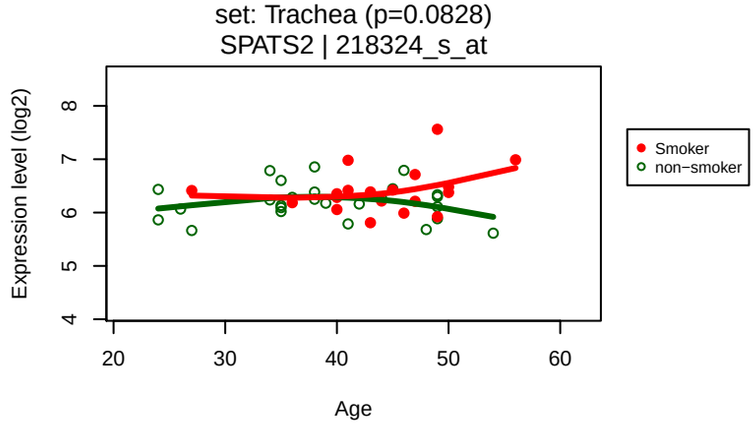

Gene expression ratio in Males
gene: SPATS2 | 218324_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.083
p=0.1448
p=0.3103
p=0.3637
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATS2 | 218324_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7833
p=0.6231
p=0.0397
p=0.9622
SmallAirway2
LargeAirway2
Trachea
Macrophage2

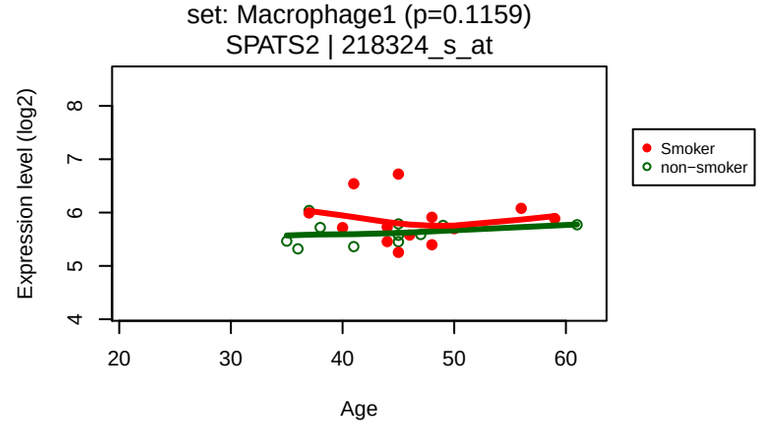

set: SmallAirway1 (p=0.1537)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker

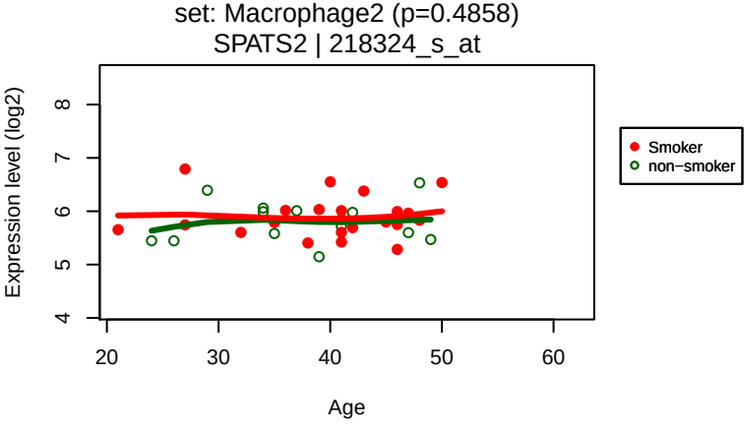

set: SmallAirway2 (p=0.1138)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker

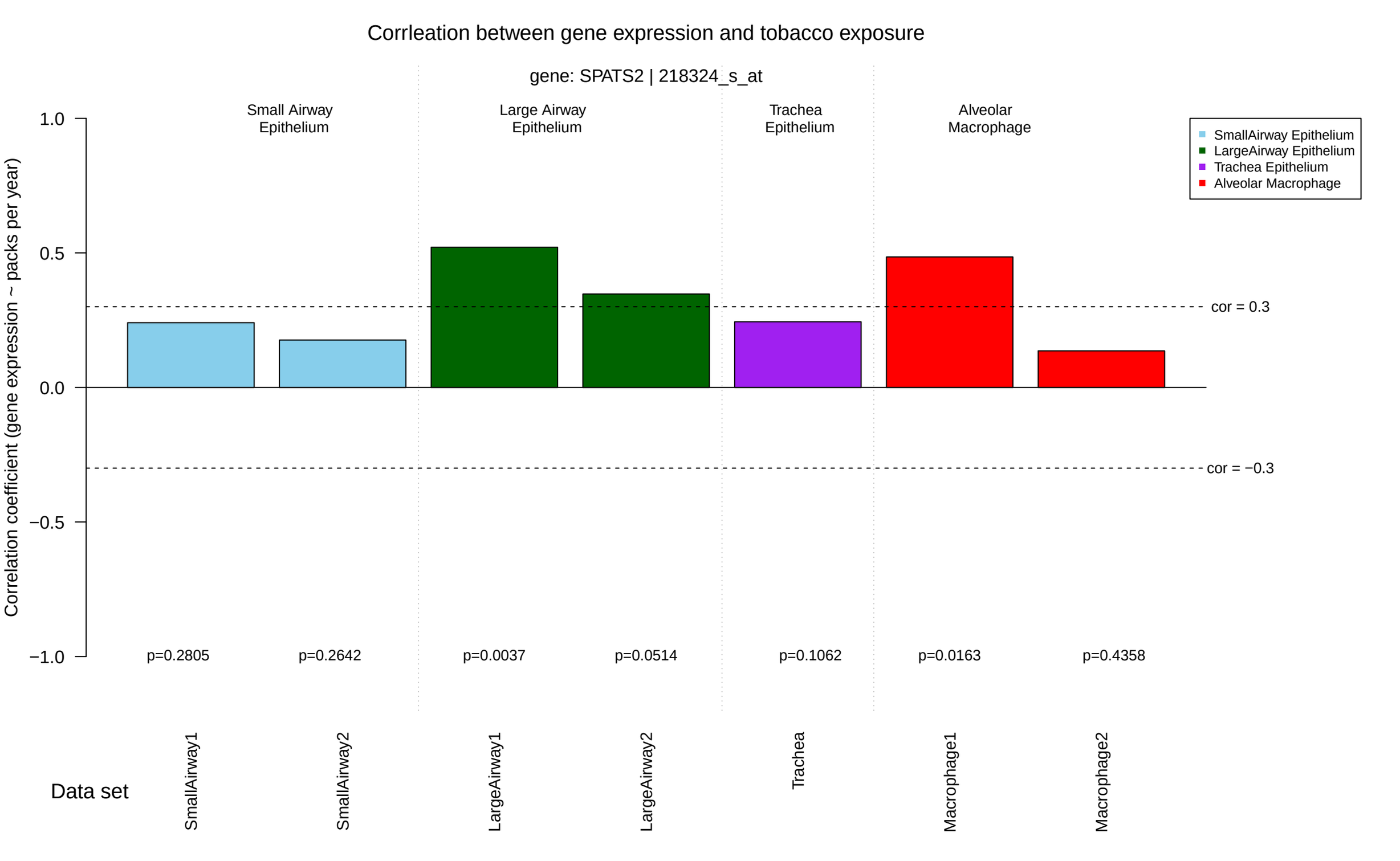

set: LargeAirway1 (p=0.0484)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4386)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0828)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1159)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4858)
SPATS2 | 218324_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATS2 | 218324_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2805
p=0.2642
p=0.0037
p=0.0514
p=0.1062
p=0.0163
p=0.4358
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2